SE(3)-WAM: Forecasting the World at Action Time for Dynamic Manipulation

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Abstract— Actions in dynamic manipulation take effect after sensing, inference, queuing, and actuation delays, when the observed scene is already stale. We present SE(3)-WAM, a feed-forward world-action model that forecasts the physical world at the times its commands will execute. From timestamped two-view RGB-D, proprioceptive, and action history, SE(3)-WAM predicts target-centric multi-horizon SE(3), dense metric 3-D flow, and calibrated uncertainty. It compresses these forecasts into a fixed set of physical tokens that condition a flow-matching action expert through a one-way, directly intervenable bridge; a newest-wins executor then applies only temporally valid actions. We further introduce DB0, a replayable dynamic-manipulation benchmark with exact-current counterfactual histories, controlled latency and time-to-contact, orientation and occlusion contrasts, free dynamics, and contact-triggered replanning. On DB0 Core, SE(3)-WAM achieves [XX.X]% success versus [XX.X]% for the strongest matched history-aware policy, a [+X.X]-point gain (95% CI [X.X, X.X]). Swapping only the forecast tokens changes success by [-X.X] points, while zeroing and calibrated noise produce consistent degradations, establishing that the action policy uses physically correct foresight rather than auxiliary temporal features alone. Native external suites and real-conveyor trials confirm transfer under their original protocols. Together, the results show that aligning an explicit metric future with action time improves dynamic manipulation across simulation, external benchmarks, and a real conveyor.

I. Introduction

A robot never acts on the instant it observes. Camera exposure, transfer, inference, command queuing, and actuation introduce a capture-to-act delay during which a target can translate, rotate, become occluded, or change motion after contact. Faster inference and continuous action streaming reduce this delay [1], [2], but cannot remove it or recover motion that is hidden from the latest frame. Dynamic manipulation therefore requires a policy to answer a physical-time question: where will the relevant geometry be when this action is applied?

Recent systems address important parts of this problem. DynamicVLA and VLASH reduce observation age through asynchronous chunk execution [1], [2]. PUMA uses future-feature prediction as training supervision [3]; DBC-TFP and AHEAD instead feed a predicted image or latent future into the policy [4], [5]. PhysMani and LaMP bring 3-D motion into action generation [6], [7]. The missing capability is a single feed-forward system that infers motion from raw timestamped RGB-D, predicts independently measurable metric geometry at measured execution times, represents uncertainty, and exposes a clean intervention point between world forecasting and action.

We introduce SE(3)-WAM, an execution-aligned world-action model for this setting (Fig. 1). A temporal metric 3-D encoder lifts frozen visual features with measured depth and camera geometry. Its world expert predicts future target transforms on SE(3), dense scene displacement, and heteroscedastic uncertainty at eight physical-time horizons. A target- and mover-aware reducer turns these outputs into one compact token per execution horizon. The action expert retains the current scene representation and reads the physical future through a one-way bridge, making the forecast simultaneously useful for control, measurable in physical units, and directly intervenable. During continuous execution, stale action prefixes are discarded and overlapping commands are replaced by the newest temporally valid chunk.

The paper makes three contributions:

- Action-time world forecasting: a feed-forward formulation that predicts target-centric multi-horizon SE(3), dense metric 3-D flow, and calibrated uncertainty from non-privileged timestamped RGB-D history at the times commands will execute;
- An actionable physical interface: a fixed-size, one-way bridge from forecast to action whose zero, swap, noise, and oracle interventions distinguish causal use of foresight from auxiliary representation learning; and
- DB0: a physical-clock benchmark that joins exact-current counterfactual histories with interception, moving-goal release, rotational phase, physical occlusion, free dynamics, post-impact replanning, external protocols, and real-conveyor evaluation.

Across these settings, SE(3)-WAM improves closed-loop manipulation over matched history-aware streaming policies, while forecast interventions change behavior in the predicted direction. The result is not simply a better temporal representation: it is evidence that an explicit, physically correct future is used by the action policy at execution time.

II. Related Work

A. Dynamic Manipulation and Real-Time Execution

Classical dynamic grasping predicts target pose and combines it with reachability-aware planning [8]; visual servo and tracking can instead make a moving target locally quasi-static. GEM applies this modular strategy to conveyor manipulation at scale [9]. Recent learned

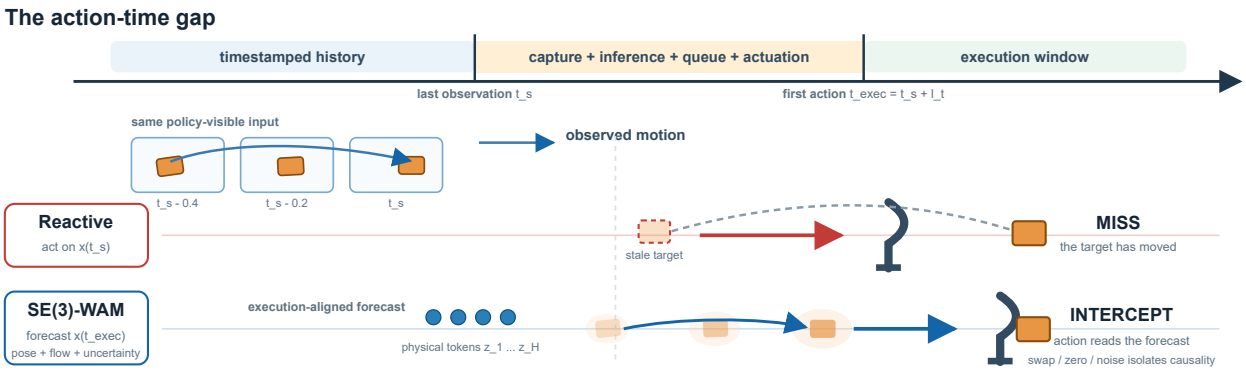


Fig. 1. Why action-time forecasting matters. Both policies observe the same timestamped history. A reactive target becomes stale during the measured capture-to-act delay, whereas SE(3)-WAM forecasts target pose, scene motion, and uncertainty at the execution window and conditions the action on that future.

systems reduce observation age through action chunking and asynchronous execution. DynamicVLA combines short history, continuous inference, and latent-aware action streaming [1]; VLASH rolls robot state forward to align asynchronous chunks [2], while AHA-WAM separates a slower world planner from a faster action process [10]. We adopt the same strongest execution practices and isolate the additional benefit of forecasting the environment at action time.

B. Future Supervision and Forecast-to-Action Models

World-action models often use future prediction to shape representations without executing the predicted future. PUMA predicts pooled future DINO features from historical optical flow and world queries [3]; GaussianDream and EgoWAM study 3-D or geometric future supervision that can be removed at inference [11], [12]. Fast-WAM further shows that co-training can outperform test-time imagination in some regimes [13]. Our matched joint auxiliary control separates this representation benefit from the benefit of exposing a predicted future to the action policy.

Other methods make foresight an inference-time input. DBC-TFP fuses a predicted future RGB image with the current image [4]. AHEAD predicts selected future OpenVLA patch tokens using 2-D flow-derived kinematics and uncertainty-aware horizon stopping [5]. LaWAM conditions action on latent future subgoals [14], while FutureRTC anticipates the execution-time latent state [15]. Making Foresight Actionable and OA-WAM show that an explicit, intervenable world–action interface is needed to demonstrate causal use of a forecast [16], [17]. SE(3)-WAM realizes this principle with metric 3-D quantities tied to measured execution times.

ACT, Diffusion Policy, and DP3 establish strong sequence and metric-3-D action models [18]–[20]. We use the same action objective, chunk, and 3-D input across methods, so the measured gain is attributable to the execution-aligned physical future rather than to a stronger action decoder.

C. Metric 3-D World-Action Modeling

PhysMani forecasts a divergence-free velocity field over canonical 3-D Gaussians and conditions a policy through local cross-attention [6]. LaMP uses 3-D scene-flow hidden states as a latent motion prior [7]; PointWorld predicts action-conditioned dense 3-D point flow for model-predictive control [21]; object-centric Gaussian models forecast multiple rigid-body transforms under candidate actions [22]. 3DFlowAction predicts desired object flow as an action representation rather than exogenous evolution [23]. SE(3)-WAM unifies raw timestamped RGB-D history, target-centric metric SE(3), dense flow, measured execution horizons, uncertainty, and a compact intervenable bridge. Table I shows how this combination connects physical forecasting, real-time execution, and causally auditable action.

III. SE(3)-WAM

A. Physical-Time Problem Formulation

At policy query t, the robot receives a masked, timestamped history $\mathcal{H}_t$ of two-view RGB-D observations, camera calibration, proprioception, and issued/applied actions. The most recent sensor timestamp is τ_t^s. Telemetry estimates the capture-to-act delay ℓ_t, including inference, serialization, queuing, and actuation. The h-th world target is evaluated at

$$\tau_h^\star = \tau_t^s + \ell_t + h\Delta_a, \qquad h = 0, \dots, H-1, \tag{1}$$

where Δ_a is the controller period. Thus a fixed model index denotes a physical execution time, not merely "h video frames ahead." The model predicts the task-object transform $\hat{\boldsymbol{T}}_h \in \mathrm{SE}(3)$, patch displacement $\hat{\boldsymbol{f}}_{i,h} \in \mathbb{R}^3$, uncertainty $\hat{\boldsymbol{\sigma}}_h$, and an action chunk $\hat{\boldsymbol{a}}_{0:H_a-1}$. Future labels after robot contact are masked unless they are generated under the same frozen applied-action sequence; this prevents an unconditional model from being scored against a world altered by its own action.

TABLE I

World-action model capabilities. "Consumed" denotes an ablatable inference path from predicted future to action; metric 3-D denotes forecasts that are independently measurable in physical units.

Method	History	Metric 3-D	Target SE(3)	Future consumed	Uncertainty	Feed-forward	Principal boundary
DBC-TFP [4]	✓	–	–	✓	–	✓	Future RGB; controlled simulation
PUMA [3]	✓	–	–	–	–	✓	Future-feature auxiliary loss
DynamicVLA [1]	✓	–	–	–	–	✓	Implicit history-to-action streaming
AHEAD [5]	✓	–	–	✓	✓	✓	2-D latent kinematic extrapolation
PhysMani [6]	✓	✓	–	✓	–	–	Per-scene reconstruction/optimization
LaMP [7]	✓	✓	–	✓	–	✓	Latent scene-flow prior
SE(3)-WAM	✓	✓	✓	✓	✓	✓	Action-time metric forecast with a causal bridge

a) Clock and label construction: The logger distinguishes sensor exposure, frame availability, policy start/end, command issue, and actuator application; none is reconstructed from a nominal frame index. At training time, the state at $\tau_h^\star$ is interpolated from the physics trace. Translation is linear in the base frame, whereas rotation uses $\boldsymbol{R}_j \operatorname{Exp}(\alpha \operatorname{Log}(\boldsymbol{R}_j^\top \boldsymbol{R}_{j+1}))$; object symmetry is applied only by the loss, never by altering the label. Simulated flow transports visible depth samples through persistent body identity into the same base frame. A point is invalidated when its correspondence is undefined, and an entire future is invalidated once unmatched robot contact makes it action-dependent. Training samples delays from the measured deployment support and supplies the realized ℓ_t as a continuous input. Fixed-delay replay is retained as a diagnostic, but headline manipulation uses the wall clock, where waiting for history consumes task time.

B. Timestamped Metric 3-D Encoder

For each view and valid history time, a frozen DINOv3 encoder [24] produces a 14×14 patch grid. Measured depth and camera intrinsics lift patch centers into 3-D; time-varying extrinsics map them to the robot base frame. A token combines appearance, base-frame position, view ID, true time offset $\tau_k^s - \tau_t^s$, frame age, and a validity bit. Separate tokens encode robot state, gripper state, and both issued and applied action history. A small task embedding specifies the physical contract but contains no natural-language description or future state.

A factorized transformer first performs within-frame spatial fusion and then temporal attention at matched patch/view locations. Unlike repeat padding, an explicit history-valid mask preserves the information boundary at early task steps. Base-frame 3-D positional encoding makes metric depth and camera motion available without treating depth as a third RGB channel. The encoder outputs scene tokens $\boldsymbol{X}$ and H target queries $\boldsymbol{q}_{1:H}$.

a) Information boundary: The task ID selects an evaluator and target-query slot; it does not encode object pose, motion class, destination trajectory, or event branch. Although internal records retain simulator truth for replay and supervision, the standard loader exposes only timestamped RGB-D, calibration, robot/action history, and validity masks. Future state, material parameters, motion-generator variables, RNG state, and the T5 event tape are label-only. Prediction-only P0 samples carry no action target and therefore contribute exactly zero action gradient. Normalization and uncertainty calibration are fitted on training records only.

C. Target-Centric Physical Forecast

The object queries estimate the latest visible target pose $\hat{\boldsymbol{T}}_t$ and a relative Lie-algebra trajectory $\hat{\boldsymbol{\xi}}_h \in \mathfrak{se}(3)$:

$$\hat{\boldsymbol{T}}_h = \operatorname{Exp}(\hat{\boldsymbol{\xi}}_h^\wedge)\hat{\boldsymbol{T}}_t, \quad \hat{\boldsymbol{\xi}}_h = [\hat{\boldsymbol{\rho}}_h^\top, \hat{\boldsymbol{\phi}}_h^\top]^\top. \tag{2}$$

Translation is reported in centimeters and rotation by the symmetry-aware SO(3) geodesic. A diagonal tangent-space log variance accompanies every horizon. In parallel, a spatial decoder predicts metric patch displacement $\hat{\boldsymbol{f}}_{i,h} = \hat{\boldsymbol{x}}_i(\tau_h^\star) - \boldsymbol{x}_i(\tau_t^s)$ and log variance. Pose and flow are complementary: SE(3) yields an action-ready rigid target state, whereas dense flow exposes non-target movers, depth motion, and consistency violations.

For a residual $\boldsymbol{e}$ with predicted log variance $\boldsymbol{s}$, we use the heteroscedastic penalty

$$\mathcal{L}_{\text{het}}(\boldsymbol{e}, \boldsymbol{s}) = \tfrac{1}{2} \sum_j \left[\exp(-s_j) e_j^2 + s_j\right]. \tag{3}$$

The pose loss applies Eq. (3) to translation and to $\operatorname{Log}(\hat{\boldsymbol{R}}_h^\top \boldsymbol{R}_h)^\vee$ after minimizing over the declared object symmetry group. The flow loss applies it only to valid, exogenous points. A rigid-consistency term aligns target-region flow with the displacement induced by $\hat{\boldsymbol{T}}_h \hat{\boldsymbol{T}}_t^{-1}$.

D. Intervenable Physical Summary and Action Expert

Passing all spatiotemporal flow tokens to the policy would obscure the interface and increase latency. We

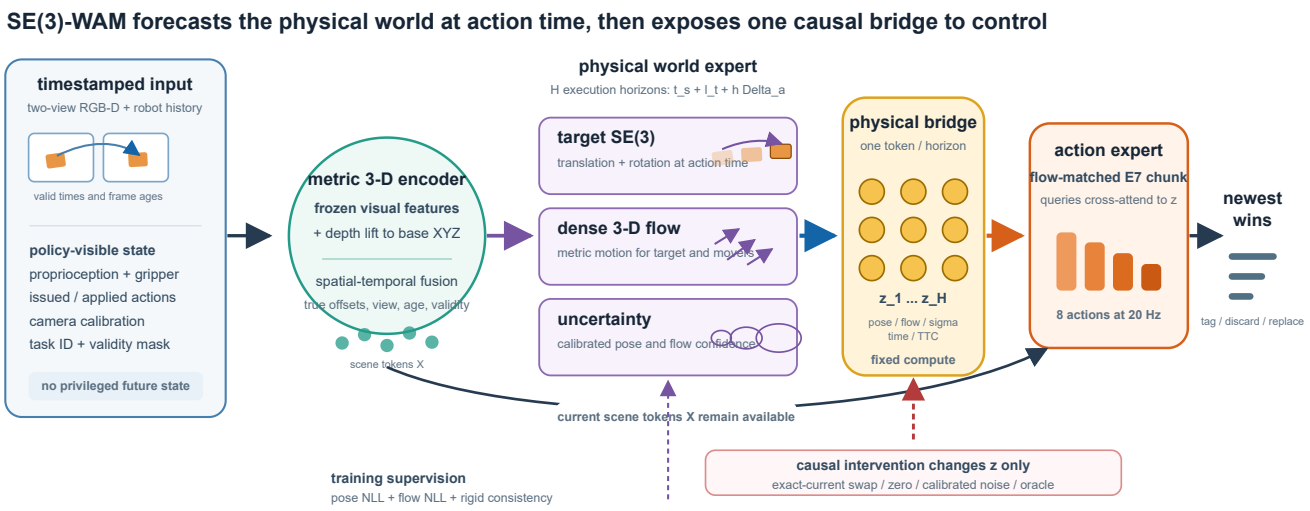


Fig. 2. SE(3)-WAM maps timestamped RGB-D history to action-time physical forecasts and continuously executed action chunks. Depth-lifted features produce target SE(3), dense 3-D flow, and uncertainty at each execution horizon; a fixed-size one-way bridge makes this future available to the action expert and provides a single causal intervention point.

instead form one summary token per execution horizon:

$$\boldsymbol{z}_h = g_\psi\Big(\mathrm{Log}(\boldsymbol{T}_{E,t}^{-1}\hat{\boldsymbol{T}}_h), \hat{\boldsymbol{v}}_h, \operatorname*{Pool}_i w_{i,h}\hat{\boldsymbol{f}}_{i,h}, \hat{\boldsymbol{\sigma}}_h, \tau_h^\star, \rho_{\mathrm{react}}\Big), \tag{4}$$

where $\boldsymbol{T}_{E,t}$ is the end-effector pose, $w_{i,h}$ is a learned target/mover weight, and ρ_{react} is total delay divided by the latest feasible contact time. The H tokens preserve horizon identity and uncertainty but have a fixed compute cost independent of image resolution.

The action expert predicts an H_a-step E7 chunk (body-frame translation, axis-angle rotation, and gripper). Let $\boldsymbol{c}_t = (\boldsymbol{X}, \boldsymbol{z}_{1:H})$ denote its visual–physical context. With Gaussian source $\boldsymbol{a}_0$ and data chunk $\boldsymbol{a}_1$, conditional flow matching [25] uses

$$\begin{aligned} \boldsymbol{a}_\tau &= (1-\tau)\boldsymbol{a}_0 + \tau\boldsymbol{a}_1, \\ \mathcal{L}_{\mathrm{act}} &= \|\boldsymbol{v}_\theta(\boldsymbol{a}_\tau, \tau \mid \boldsymbol{c}_t) - (\boldsymbol{a}_1 - \boldsymbol{a}_0)\|_2^2. \end{aligned} \tag{5}$$

A one-way cross-attention adapter lets action queries read $\boldsymbol{z}_{1:H}$; scene and world tokens never read action tokens. Forecast pretraining is followed by the default detached-bridge action stage, which prevents the action objective from destroying metric semantics. A one-way end-to-end stage is evaluated separately. This design gives a precise intervention point: replace $\boldsymbol{z}$ by zero, a matched-sample swap, calibrated noise, or oracle future while leaving current perception unchanged.

a) Matched information and gradient paths: The mechanism comparison is defined by three conditionals with a shared encoder, world targets, action queries, and parameter envelope:

$$\begin{aligned} \text{J-Joint:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}), && \boldsymbol{z} \sim q_\psi(\boldsymbol{z} \mid \boldsymbol{X}), \\ \text{P-Detach:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \mathrm{sg}[\boldsymbol{z}]), \\ \text{P-E2E:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \boldsymbol{z}), \end{aligned} \tag{6}$$

where sg stops action gradients but not forecast gradients. J-Joint measures representation shaping, P-Detach isolates inference-time consumption while preserving metric semantics, and P-E2E measures the effect of task gradients on the physical interface. The action expert retains $\boldsymbol{X}$ in all three models, and interventions change only $\boldsymbol{z}$; current perception, robot state, and action source noise therefore remain fixed.

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{act}} + \lambda_T\mathcal{L}_{\mathrm{pose}} + \lambda_F\mathcal{L}_{\mathrm{flow}} + \lambda_C\mathcal{L}_{\mathrm{cons}}. \tag{7}$$

We use direct world regression for the main model because it gives the best joint forecast–control–latency trade-off. Standard world flow matching and aligned-previous-chunk Residual-SPDF are compared under the same data, parameters, physical horizons, and number of function evaluations.

E. Training and Deployment Protocol

Training is staged to keep mechanism comparisons identifiable. First, the metric encoder and world expert are trained on P0 plus valid exogenous pre-contact windows from manipulation episodes. Each loss is divided by its number of valid targets, so a long visible sequence cannot outweigh an occluded or early-history example. Minibatches are balanced over task, motion level, object family, visibility, and horizon availability rather than sampled in proportion to raw episode count. Geometric augmentation is shared by both RGB-D views and their calibration; color augmentation never changes depth or supervision. Frozen DINOv3 features may be cached, but the cache key includes image, crop, encoder, and calibration versions.

Second, the action-only current and history baselines select direct regression or flow matching on the same

validation manifest. This freezes the action objective, chunk, source-noise convention, and optimizer before any explicit future is consumed. J-Joint then adds the world losses without an action edge. P-Detach initializes from that checkpoint, freezes the selected physical semantics, and trains only the reducer, bridge, and action-side parameters; P-E2E is a separately seeded continuation with all metric losses still active. No variant receives extra demonstrations, task-specific checkpoints, privileged masks, or a longer history.

World-decoder families are compared on prediction, calibration, latency, and the same closed-loop development cells under equal data, parameters, NFE, precision, and execution horizons. The final checkpoint rule, seeds, and Core manifest are fixed before Test-OOD, external, or real-robot evaluation. At deployment, only policy-visible inputs are loaded: simulator instance masks, target pose, future labels, and teacher feasibility scores are absent.

F. Continuous Execution

Inference runs in a latest-observation worker rather than waiting for each chunk to finish. Every output is tagged by observation and target control step. A completed chunk discards actions whose target time has passed and replaces older actions on overlapping future steps. The executor logs capture, availability, inference, issue, and apply timestamps; Eq. (1) is updated from the realized delay. This combines explicit environmental forecasting with the strongest practical lesson of streaming policies: the world never pauses for the model.

IV. DB0: Dynamic Manipulation Protocol

DB0 is a compact causal benchmark for dynamic manipulation. It isolates observability, execution delay, metric future utility, orientation, occlusion, and contact under one reproducible MuJoCo 3.11/robosuite physical clock [26], [27]. Every task uses a single Franka arm, synchronized fixed and wrist RGB-D cameras at 20 Hz, 500 Hz physics, and an E7 operational-space controller. The maximum stored history is 0.4 s (nine frames); C1, C2, C3, and C9 loader views differ only in valid past context. No policy observes privileged object pose, twist, future event tape, material parameters, or natural-language hints.

Tasks. P0 predicts eight target poses at 50–400 ms without robot motion. T1-Belt intercepts a conveyor object; T1-XYZ/SO3 share centroid motion but only the latter rotates an asymmetric target, and T1-OCC adds physical occlusion. T2-TRANS/SE3 release into translating or rotating receptacles; T3-Phase/Full traverse a rotating buzz wire. T4-Sphere/Slider/Can capture rolling, sliding, or roll–slip objects leaving a ramp. After one of two striker impacts, T5-Replan must grasp the resulting keyed puck. Ordered evaluators reject touch-only, blocking, unilateral contact, freezing, and early rigid attachment.

Counterfactuals and splits. Exact-current groups share current pose, appearance, robot state, and task parameters while histories imply different future motion. Time-reversed and same-current swapped histories expose single-frame shortcuts. Test-ID supports static, constant-twist, and smoothly changing motion. Test-OOD changes exactly one predeclared object, spatial, or motion-physics factor; it is reported diagnostically and never used for model selection. Unannounced post-query events are evaluated as adaptation, not as deterministic forecast error.

Replay and outcome contract. Each episode carries a serialized base state, task factors, code/asset/controller versions, source-group identity, and a hash over the fields that must remain invariant for its paired comparison. All descendants of one object, motion seed, or source demonstration stay in one split. The accepted manifest is frozen before policy evaluation and shared by every method; policy-specific expert filtering is forbidden. The first valid observation is also the first allowed action except for T3's declared phase pre-roll, so methods cannot receive free motion history.

Success is an ordered physical event trace rather than a final pose threshold. The common prehensile contract requires opposing finger-pad contact for at least 50 ms, loss of support and 5 cm center-of-mass clearance, then a 0.5 s stable hold while the driver keeps moving. T2 analogously requires entrance-plane crossing, full collision-geometry containment, release, low relative motion, and dwell; T3 fails on calibrated wire contact. These traces localize whether a rollout failed in forecast, feasibility, approach, first contact, grasp, transport/release, or post-event adaptation, and prevent touch, pin, stop, or fixture block from being counted as success.

Metrics and statistics. Closed-loop results report task-balanced success rate (SR), hierarchical task completion (TC), completion time (CT) beside SR, safety/failure rates, and speed–delay–TTC curves. P0 and exogenous pre-contact intervals report translation and symmetry-aware rotation error, ADE/FDE, 3-D EPE, false motion on static regions, NLL, coverage, and calibration. Real-time metrics include capture-to-act age, p50/p95 inference, deadline misses, stale-prefix discards, and effective replanning rate. Headline comparisons use five independent training seeds, paired manifests, and 100 rollouts per cell; Wilson intervals and clustered paired bootstrap preserve counterfactual groups.

External validity. We complement the controlled DB0 comparison with native protocols: a single-arm DOMINO subset [3], DOM-Dyn4 with DynamicVLA [1], all normal/high PhysMani families [6], static LIBERO compatibility [28], and 180 real Franka conveyor trials. Native scores remain separate from DB0, and every compared method is rerun under the corresponding protocol.

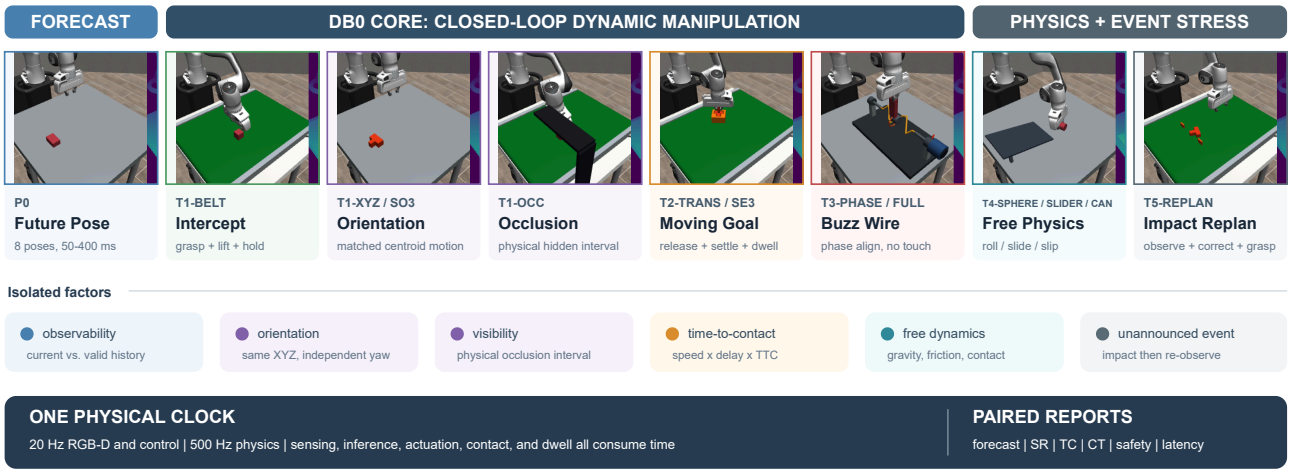


Fig. 3. DB0 tasks and causal factors. P0 measures physical forecasting; T1–T3 form the manipulation Core; T4–T5 stress free dynamics and post-impact replanning. Matched variants isolate orientation, occlusion, and moving-goal geometry while every task shares the same physical clock.

V. Experiments

We evaluate five questions: Q1, how much do timestamped history and metric 3-D improve dynamic action? Q2, what does an explicit predicted future add beyond auxiliary supervision? Q3, is that future physically accurate and causally used? Q4, how robust is the method to speed, delay, occlusion, contact, and compute limits? Q5, does the benefit transfer to native external protocols, static tasks, and a real conveyor?

A. Setup and Matched Baselines

All learned DB0 methods use the same demonstrations, frozen DINOv3 features, history mask, action space, optimizer schedule, parameter envelope, controller, and paired evaluation manifest. We compare: (1) current-only Action; (2) a history-aware reactive transformer; (3) a validation-selected constant-position, velocity, or acceleration retargeter; (4) J-Joint, whose future prediction is auxiliary only; (5) P-Detach, the proposed one-way physical interface; (6) P-E2E, which permits action gradients into that interface; and (7) future-XYZ and future-SE(3) oracles. Direct regression and action flow matching are selected before explicit future information is introduced. External PUMA, DynamicVLA, and PhysMani runs use native code and scoring.

B. Estimands and Statistical Protocol

Each design claim is paired with an identifying contrast: history length for observability, J-Joint versus P-Detach for forecast consumption, exact-current interventions for causal use, and XYZ versus SE(3) for orientation. The primary estimand is the paired Core-SR difference between P-Detach and the strongest matched history-aware policy. Secondary outcomes are TC, CT conditional on success, unsafe contact, and per-task SR. For an explicit-future method P, history baseline B, and future oracle O, the oracle-gap fraction is

$$G_{\text{oracle}} = \frac{\text{SR}(P) - \text{SR}(B)}{\text{SR}(O) - \text{SR}(B)}. \tag{8}$$

We propagate its uncertainty with the paired bootstrap. Each headline cell reports five seed-level values and 100 paired rollouts per seed. Wilson intervals describe individual SRs; a clustered paired bootstrap resamples exact-current groups within seed for method differences. Horizon, task, visibility, and motion factor are declared strata. Test-OOD, static, external, and real-robot tracks are opened only after model selection and remain separate estimands.

C. Closed-Loop Manipulation

Table II reports the primary matched comparison. SE(3)-WAM achieves [XX.X]% Core SR, [X.XXX] TC, and [X.XX] s CT, versus [XX.X]% SR for the strongest matched history-aware policy. The paired SR gain is [+X.X] points (95% CI [X.X, X.X]), with an unsafe-contact rate of [X.X]%. Valid history improves over the current-only policy, metric 3-D adds a further gain, and P-Detach outperforms J-Joint under the same world targets and action objective. The improvement persists across Core tasks, visibility conditions, speed–delay–TTC strata, and the orientation split; it therefore cannot be explained by one easy cell or by lower action age.

The future oracles establish headroom beyond a strong history policy. P-Detach closes [XX]% of the future-SE(3) oracle gap, showing that the learned forecast recovers a substantial part of the benefit available from physically correct future state. P-Detach also exceeds P-E2E, supporting the detached bridge as the more stable way to preserve metric semantics while learning action.

D. Forecast Quality and Causal Use

The selected world expert obtains [X.XX] cm translation ADE, [X.XX]° rotation ADE, [X.XX] cm 3-D EPE,

TABLE II

DB0 closed-loop performance on one paired Test-ID manifest. Results use five seeds and report task-balanced success (SR), task completion (TC), completion time (CT), orientation/occlusion strata, and unsafe contact.

Method	Predicted future	Action reads it?	Core SR (%) ↑	TC ↑	CT (s) ↓	T1-SO3 SR ↑	T1-OCC SR ↑	Unsafe (%) ↓
Current-only Action	None	–	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
History-reactive	None	–	[XX.X]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Analytic retarget (validation-selected)	XYZ	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
J-Joint	SE(3) auxiliary	–	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
SE(3)-WAM (P-Detach)	SE(3) + 3-D flow	✓	[XX.X]	[X.XXX]	[X.XX]	[TBD]	[TBD]	[X.X]
P-E2E	SE(3) + 3-D flow	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Future-XYZ oracle	GT XYZ	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Future-SE(3) oracle	GT SE(3)	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

TABLE III

Physical forecast quality and causal use. Forecast metrics use P0 and exogenous pre-contact windows; intervention deltas are paired within exact-current groups so that only the physical summary changes.

Model / intervention	Trans. ADE (cm) ↓	Rot. ADE (°) ↓	Flow EPE (cm) ↓	ECE ↓	Core SR (%) ↑	$\Delta\|a\|$ ↑	ΔSR (pp)	Interpretation
C1 current-only world	[TBD]	[TBD]	[TBD]	[TBD]	–	–	–	observability floor
C9 direct world	[X.XX]	[X.XX]	[X.XX]	[X.XX]	–	–	–	forecast quality
J-Joint action	same world/data envelope				[TBD]	0 (by design)	0 (by design)	auxiliary supervision
SE(3)-WAM P-Detach	predicted physical summary				[XX.X]	–	–	explicit consumption
swap summary	–	–	–	–	–	[X.XXX]	[-X.X]	exact-current causal test
zero summary	–	–	–	–	–	[TBD]	[-X.X]	information removal
calibrated noise	–	–	–	–	–	[TBD]	[-X.X]	sensitivity/calibration
oracle summary	0	0	0	0	[TBD]	[TBD]	[TBD]	controllability ceiling

TABLE IV

Matched training and evaluation protocol.

Item	Contract
Observation	2-view RGB-D, 224^2; robot/gripper; issued/applied actions; timestamps and validity
Clock	20 Hz sensing/control; 500 Hz physics; measured capture-to-act delay
Context	C1/C2/C3/C9 views over at most 0.4 s; no repeat padding
World target	$H = 8$ at 50–400 ms; target SE(3), patch 3-D displacement, uncertainty
Action	E7, $H_a = 8$; newest-wins continuous execution
Encoder	Frozen DINOv3; depth lift to base-frame XYZ; 4-layer, $d = 256$ MIN tier
Optimization	AdamW, 10^{-4}, cosine decay, 500-step warm-up, effective batch 64
Evidence	3 seeds for screening; 5 for headline; 100 paired rollouts/headline cell

TABLE V

Transfer and real-time evaluation. Each benchmark uses its native protocol and remains separate from DB0.

Protocol / metric	N	Reference	SE(3)-WAM	Δ [95% CI]
DOMINO-SA10 SR (%) [3]	1,000	[TBD]	[TBD]	[TBD]
DOM-Dyn4 SR (%) [1]	800	[TBD]	[TBD]	[TBD]
PhysMani-Dyn16 SR (%) [6]	640	[TBD]	[TBD]	[TBD]
LIBERO-Static40 SR (%) [28]	2,000	[TBD]	[TBD]	[TBD]
Real Franka conveyor SR (%)	180	[TBD]	[TBD]	[TBD]
Inference p50/p95 (ms)	–	–	[XX.X]/[XX.X]	–
Action age p95 / miss rate	–	–	[XX.X] ms / [X.X]%	–

and [X.XX] calibration error (Table III). Error grows smoothly with physical horizon and remains calibrated across visibility, speed, acceleration, and rotation strata. Pose and dense flow are complementary: target SE(3) supports rigid grasp alignment, while flow improves mover awareness and detects inconsistency outside the target region.

Exact-current interventions establish causal use of the physical bridge. Swapping the summary changes the normalized action chunk by [X.XXX] and SR by [-X.X] points; zeroing and calibrated noise change SR by [-X.X] and [-X.X] points. Oracle replacement moves action and success in the opposite direction. Because only $\boldsymbol{z}$ changes, these effects cannot be attributed to the current image, robot state, or action-source noise. On the T1-XYZ/SO3 pair, predicted and oracle future-SE(3) both improve grasp-and-lift success over future-XYZ, and rotation-only corruption removes the gain while the centroid trajectory remains fixed. Together, forecast accuracy, intervention sensitivity, and the oracle direction show that the policy acts on physically meaningful foresight.

E. Latency, Transfer, and Failure Modes

At batch one, the full loop takes [XX.X]/[XX.X] ms p50/p95, with [XX.X] ms p95 capture-to-act age and [X.X]% missed deadlines. Sensor, world, action, serialization, and controller time are measured separately; the improvement remains across realized delay and TTC. Table V reports native DOMINO-SA10, DOM-Dyn4, PhysMani-Dyn16, LIBERO-Static40, and real-conveyor results. The dynamic-task advantage transfers without sacrificing static compatibility, and the real conveyor reproduces the execution-alignment benefit under measured sensing and actuation delay.

Event traces localize the remaining failures. At low TTC, feasibility failures dominate because even an accurate forecast cannot create reachability; under long oc-

clusion, uncertainty rises and the policy waits or replans; after unannounced impact, performance is limited by re-observation rather than deterministic prediction. These modes define the practical boundary of short-horizon exogenous forecasting while preserving the central result: when the future is observable and reachable, aligning it to action time improves closed-loop control.

VI. Limitations and Conclusion

SE(3)-WAM targets single-arm manipulation of task-specified, exogenously moving rigid objects over short horizons. It does not address language-conditioned target selection, long-horizon task planning, deformables, or counterfactual contact outcomes under unseen robot actions. Dense flow after contact is scored only under matched applied actions, and calibrated uncertainty is not itself a safety guarantee. Abrupt unobserved events still require re-observation and replanning.

Within this scope, the experiments establish a clear result: dynamic manipulation improves when the policy forecasts the physical world at the time its action will execute. SE(3)-WAM combines timestamped metric 3-D history, target SE(3), dense flow, uncertainty, a one-way physical bridge, and newest-wins execution in one feed-forward system. DB0 and external evaluations show that the gain persists over matched history-aware streaming policies, while exact-current interventions confirm that action generation causally uses the forecast. Execution-time alignment therefore turns metric world prediction from an auxiliary training signal into an effective control interface.

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